Original Article

Anterior Cruciate Ligament Tear Location in National Football League Athletes Based on Magnetic Resonance Imaging

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Abstract

Background: Anterior cruciate ligament (ACL) tears are common and often devastating injuries among elite athletes. While interest in ACL preservation has highlighted the importance of ACL tear location to treatment decision-making, there is limited data on ACL tear location in elite athletes.

Purpose: We sought to describe the distribution of ACL tear location among National Football League (NFL) athletes and test the hypothesis that tear location is associated with injury mechanism.

Methods: Using a dataset from a study published in 2023, we identified 191 players with magnetic resonance imaging (MRI) of acute ACL tears occurring in NFL practice or games from 2015 to 2019; these were reviewed by 2 musculoskeletal radiologists. Data from these MRIs were linked to each player's electronic medical record data, and injuries from 2018 to 2019 were linked with available video to assess for an association between tear location and mechanism of injury in 73 of those players.

Results: Overall, 3% of tears were type I, 20% type II, 74% type III, 2% type IV, and <1% type V. For noncontact injuries, 71% were type III and 29% type II. Among indirect contact injuries, 78% were type III, 13% type II, 4% type IV, and 4% type V. In direct contact injuries, tears were 73% type III, 13% type II, and 13% type IV.

Conclusion: This retrospective review of MRI and video data of ACL tears in NFL players found that mid-substance tears were the most common. There was a higher proportion of type III tears and a lower proportion of type I tears compared to prior studies, suggesting these athletes may be less amenable to preservation techniques.

Level of Evidence: Level IV: cross-sectional study.

Keywords

ACL, sports, imaging, ligament repair, knee

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Introduction

Athletes in the National Football League (NFL) are at risk for anterior cruciate ligament (ACL) injury.[1] Historically, these injuries have been treated with surgical reconstruction. However, there is recent interest in using ACL-preserving techniques, such as ACL repair, that offer possible benefits such as decreased donor site morbidity while maintaining proprioceptive function and native ACL kinematics.[2-5] The potential of this approach depends on the ACL injury location. In an early study assessing the outcomes of primary ACL repairs based on tear location, the authors reported better results in proximal, type I avulsion tears with minimal tissue left on the femur.[6] More recent data have suggested that primary repair of proximal ACL tears may have similar failure rates compared to reconstruction.[7] While ACL preservation will not replace reconstruction, it may be a consideration in a subset of patients with type I tears. As a result, it is important to understand the percentage of the population that may be eligible. Using magnetic resonance imaging (MRI), recent studies have reported a prevalence of type I tears ranging from 4% to 26%.[6,8-19] Patients in these studies were retrospectively selected from a general population with ACL tears.

Factors including age, body mass index (BMI), and injury mechanism are suspected to influence ACL tear location.[14,15,20] One study found that type I tears were more common in people older than 35 years, but no difference in correlation was noted between mechanisms of injury involving skiing, soccer, or basketball.[14] However, to our knowledge, the exact injury mechanism, such as contact or noncontact, has not been studied regarding tear location. We are also not aware of any published data regarding the location of ACL tears in elite or professional athletes. Therefore, we sought to describe the distribution of ACL tear location among NFL players. We hypothesized that mid-substance tears would be the most common tear location in this population and that, in these athletes, the location of ACL tears would be associated with the mechanism of injury.

Methods

This study was approved by the Mount Sinai School of Medicine Institutional Review Board (IRB) (#0006018123) and through the NFL-NFL Players Association Player Scientific and Medical Research Approval Process.[21] It is based on the same dataset we developed in a 2023 investigation reporting concomitant intra-articular injuries associated with ACL tears.[22] As previously described, 314 ACL tears occurred during the 2015 through 2019 NFL seasons. Images of 294 (94%) of these tears were uploaded to the NFL-wide imaging system (Infinitt), of which 274 (87%) MRIs from the time of injury were identified. Of these, 208 knees (66%) had complete MRIs of the ACL tear, but 17 players (5%) were excluded because of evidence of prior ACL reconstruction. As a result, 191 players (61%) were analyzed and included in the study. MRIs were independently reviewed by 2 fellowship-trained musculoskeletal radiologists with 11 and 20 years of experience, respectively. For each MRI, the location (proximal: type I, proximal/mid: type II, mid: type III, mid/distal: type IV, or distal: type V) of ACL tear or ACL graft tear was noted based on a previously described classification scheme.[14]

After the MRIs were analyzed, 2 independent analysts queried the radiologists to clarify any unclear, inconsistent, or missing data. Imaging reads by each radiologist were used to assess inter-rater reliability. Any discrepant findings were resolved by consensus between the radiologists to provide a single data point for the analysis. For ACL tears that occurred in 2018 and 2019, imaging data were linked to a previously reported video review analysis[23] to determine the association between injury mechanism and tear location on MRI. If the injury mechanism could not be identified through video, it was extracted from the electronic health record (EHR) as reported by medical staff. Prior to the 2018 season, injury mechanism reporting in the EHR was changed to include 4 types of contact: noncontact, indirect contact (contact to another body part above or below the knee), direct contact (contact to the knee), and other or unknown. Due to this change in reporting, analysis of the agreement between the mechanism of injury and imaging findings is not available prior to 2018. Therefore, the analysis between injury mechanism and ACL tear location includes only injuries from 2018 to 2019, comprising 73 players from the larger cohort. In the previous analysis, there was good agreement between the EHR-reported contact type and video-identified contact type.[23]

Descriptive statistics were calculated, including counts, means, standard deviations, medians, interquartile ranges, and ranges. κ statistics were calculated to assess inter-rater reliability between radiologists, where above 0.90 indicated almost perfect agreement, 0.80 to 0.90 indicated strong agreement, 0.60 to 0.79 indicated moderate agreement, 0.40 to 0.59 indicated weak agreement, 0.21 to 0.39 indicated minimal agreement, and 0 to 0.20 indicated no agreement. κ statistics were averaged across all data points collected to determine overall agreement. Injury mechanism and ACL tear location on MRI were compared using Fisher's exact χ^2 tests to test a difference in the proportion of tear location by contact type. A significance level of 95% was set a priori. The Strengthening the Reporting of Observational Studies in Epidemiology checklist was used for reporting this cross-sectional study.

Results

Type I tears were identified in 3% of players, type II tears in 20%, type III tears in 74%, type IV tears in 2%, and type V tears in <1% of players (Table 1). Inter-rater reliability between the 2 musculoskeletal radiologists was minimal, with a κ value of 0.37. All discrepancies (n=20) were within ±1 type, that is, type II or type III rather than type II

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or type IV; these were resolved by consensus between the radiologists and included in the analysis.

The mechanism of ACL tear was evaluated from a group of 73 players in the 2018 to 2019 cohort. Noncontact ACL tears were seen in 47% (n=34) of players, indirect contact ACL tears in 32% (n=23) of players, and direct contact ACL tears in 21% of players (n=15). There was 1 (1%) ACL tear in this group with an unknown or missing mechanism. In players with noncontact injuries, 29% of tears were type II and 71% were type III (Table 2). In those with indirect contact injuries, tears were: 13% type II, 78% type III, 4% type IV, and 4% type V. Lastly, in players with direct contact injuries, tears were 13% type II, 73% type III, and 13% type IV.

Table 1. Anterior Cruciate Ligament Tear Location on Magnetic Resonance Imaging, National Football League, 2015 to 2019 (n=191).

Tear location	N	%	Kappa statistic
Proximal avulsion/type I	6	3%	0.37
Proximal/type II	39	20%	
Mid-substance/type III	141	74%	
Distal/type IV	4	2%	
Distal avulsion/type V	1	<1%	

Discussion

Our study findings present new information on the type and location of ACL tears in NFL athletes and the association between tear location and injury mechanism. In the MRI analysis, type III tears in the ligament mid-substance were the most common pattern (74%), followed by type II proximal tears (20%). In the video analysis, type II tears were more frequently associated with noncontact mechanisms (67%) compared to indirect contact (20%) and direct contact (13%), although there was no statistically significant difference in the distribution of tear types by injury mechanism. Type III tears were the most common in each mechanism group. Type IV tears were more prevalent with indirect (33%) and direct contact (67%) mechanisms. There were no distal type IV or V tears observed resulting from noncontact, 2 distal type IV tears observed with direct contact, and 1 distal type IV as well as 1 distal type V tear observed as a result of indirect contact.

This study had multiple limitations. MRIs were performed at several different centers with different imaging protocols; this may have affected reliability in assessing tear location and ultimately factored into minimal interrater agreement. Additionally, the true tear location was not known, as surgical data were not available to confirm the MRI findings. Currently, there is limited evidence assessing and comparing MRI tear location to arthroscopic tear location, with recent studies reporting diagnostic accuracy ranging from 35% to 70%.[16,24,25] Assessment of tissue quality is also a consideration in preservation, but such evaluation was outside of the scope of this study. Mechanism of injury was reported as noncontact, indirect contact, or direct contact, but a specific mechanism, such as valgus or hyperextension injury, was not recorded. Furthermore, the analysis was limited by the number of players with a mechanism of injury available (n=73). Stratification of tear location by specific age, BMI, or position was not performed.

Our findings suggest that the majority of ACL tears in NFL athletes are mid-substance tears. This is also true in the general population. However, the proportion of type III tears in this study was slightly increased. In our study, 74% of ACL tears were mid-substance compared to prior studies that reported a range of 44% to 62%.[6,8-18] In addition, the majority of ACL tears resulted from noncontact mechanisms. There was no statistically significant difference in the distribution of tear types by injury mechanism. In comparison to prior studies, there was a much lower prevalence of type I tears in our population, present in only 3% of ACL injuries, compared to a previously reported prevalence ranging from 4% to 26%.[6,8-19]

There are several possible explanations for the different distribution of ACL tear locations in our cohort compared to previous studies, including age, intrinsic factors associated with this population, and energy associated with injury occurrence. In 1 study performed by van der List et al, type I tears were significantly more frequent in patients older than 35 years compared to patients aged 18 to 35 years.[14] Furthermore, type III tears have been more associated with younger adults and pediatric populations, particularly in individuals older than 14 years.[10,14,15] Since the average age of players in this study is around 25 years,[1] our findings are in agreement with prior data suggesting that younger patients may be more likely to sustain a mid-substance tear. In relation to energy of injury, it has been theorized that proximal avulsion injuries are more common in lower energy injuries, whereas the ligament is more commonly disrupted in the mid-substance in higher-energy injuries.[14] Compared to the general population, ACL injuries in NFL players are more likely to be higher-energy injuries, given the contact nature of American football. Our findings of increased type III tears and decreased type I tears align with this theory. Our findings could also be a result of selection bias, as previous studies included cohorts with a high prevalence of patients undergoing repair.[8,12-16] Another possible explanation is observer bias, as we had 2 musculoskeletal radiologists review the images, whereas previous studies used a combination of musculoskeletal radiologists, orthopedic surgeons, and research fellows.[14,15]

Assignment of precise tear location using this classification system was challenging, as reflected in the low interobserver agreement between the readers. One major obstacle is the MRI appearance of the acutely torn ACL. The majority of MRI exams were performed shortly after injury, when

Table 2. Anterior Cruciate Ligament Mechanism on Video Analysis Matched With Tear Location on Magnetic Resonance Imaging, National Football League, 2018 to 2019 (n = 73).

	Noncontact	Indirect contact	Direct contact		Other/unknown/missing
	N (%)	N (%)	N (%)		N (%)
Tear location	34 (47%)	23 (32%)	15 (21%)	*P* value	1 (1%)
Proximal avulsion/type I (N = 0)	0 (0%)	0 (0%)	0 (0%)	.14	0 (0%)
Proximal/type II (N = 15)	10 (29%)	3 (13%)	2 (13%)		0 (0%)
Mid-substance/type III (N = 54)	24 (71%)	18 (78%)	11 (73%)		1 (100%)
Distal/type IV (N = 3)	0 (0%)	1 (4%)	2 (13%)		0 (0%)
Distal avulsion/type V (N = 1)	0 (0%)	1 (4%)	0 (0%)		0 (0%)

the torn ligament fibers often have a disorganized and edematous appearance. This can mask the exact location of the fiber disruption. A wide range of MRI findings described for the torn ACL include a mass-like appearance replacing the normal ACL in the intercondylar notch, ill-defined fiber bundles oriented at an angle shallower than the Blumensaat line, an empty appearance of the lateral sidewall of the intercondylar notch, and poorly seen ACL bundles. In addition, even in cases with a clearly seen tear cleft across the ACL fibers, exact tear location may be difficult to assign, given the oblique course normally taken by the ACL, which does not conform to the standard sagittal, coronal, and axial MRI planes. The MRI studies were also obtained across a 5-year period at multiple imaging sites, using different MRI systems with variable field strength, coil selection, imaging planes, and pulse sequences. The low agreement in this study by 2 experienced subspecialty MSK radiologists familiar with the classification system raises the question of the system's accuracy for determining precise tear location when applied to routine MRI examinations, given these limitations. Some MRI protocols contain an additional sequence obtained in the oblique sagittal plane of the ACL. This sequence shows the ACL in its entire long axis and may allow more accurate determination of tear location, but it is not routinely performed during knee MRI.

Regardless of the challenges with assessing the location of ACL tears on MRI, ACL preservation can only be considered in a subset of patients based on the location of the ACL tear. While type I tears are the most amenable to preservation, some have proposed a role for preservation with type II proximal tears with augmentation, such as internal bracing rather than primary repair alone.[26] While these injuries might be candidates for augmented repairs, NFL athletes have an increased distribution of type III tears and decreased type I tears overall, which suggests they are less likely to benefit from preservation techniques and that ACL reconstruction is likely to remain the gold standard.

In conclusion, this is the first study we are aware of reporting on the distribution of ACL tear location in professional American football athletes based on MRI analysis. Type III mid-substance tears were the most common, followed by type II proximal tears, with very few type IV distal tears, proximal avulsion tears, and type V distal avulsion tears. There was an overall increase in type III tears and a decrease in type I tears compared to data previously seen in the general population. These findings suggest that very few of these athletes may be candidates for ACL preservation techniques; however, further research is needed, including intraoperative confirmation of tear location and larger cohorts including athletes from other sports.

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Ethical Considerations

All procedures followed were in accordance with the ethical standards of the responsible committee on human experimentation (institutional and national) and with the Helsinki Declaration.

Consent to Participate

Informed consent was waived for all participants included in this study by the IRB at Mt. Sinai School of Medicine.

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Declaration of Conflicting Interests

The authors declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: Robert Brophy, MD, reports being Chair of the National Football League (NFL) Musculoskeletal Committee. Ed Wojtys, MD, reports relationships with University of Michigan, the National Institutes of Health, the National Institute of Arthritis and Musculoskeletal and Skin Diseases, and Sports Health Journal. Ben Stollberg, MS, ATC, reports being a paid consultant for the NFL during the writing of this article. Christina Mack, PhD, is a salaried employee of IQVIA, which is a paid research consultancy with the NFL, and also reports

relationships with AZ, J&J, and MindMed. Mackenzie Herzog, PhD, is a salaried employee of the NFL. The other authors declare no potential conflicts of interest.

Required Author Forms

Disclosure forms provided by the authors are available with the online version of this article as supplemental material.

Supplemental Material

Supplemental material for this article is available online.

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